Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION
AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

Name: Yhamel M. De Guzman **Section:** BSCS 1-1

Course/Program: BSCS 1-1 **Date:** October 23, 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just “programming.” As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI

Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, building, testing, and maintaining software such as websites, mobile apps, desktop programs, and games.	Software Developer, Web/Mobile App Developer, Game Developer, QA Engineer
Systems & Cloud Infrastructure	Setting up, running, and protecting the servers, networks, and cloud platforms that keep computer systems available and reliable.	System Administrator, Cloud Engineer, DevOps Engineer, Network Engineer
Data Science & AI Engineering	Collecting and analyzing data and building machine learning and AI models to find patterns and support decisions.	Data Analyst, Data Scientist, Machine Learning Engineer
Cybersecurity & Threat Defense	Protecting systems, networks, and data from attacks; detecting, investigating, and responding to security threats.	Security Analyst, SOC Analyst, Penetration Tester
Information Systems & Enterprise IT	Planning, managing, and supporting the IT systems that run an organization, such as databases, business software, and IT help desks.	Systems Analyst, IT Manager, Database Administrator, Business Analyst

My Chosen Pathway (Part B)

I am most interested in: Software Engineering (creative focus: UI/UX design, game development, and digital art tools)

Why I chose it:

I am a traditional artist with a computer science background, so I want a pathway where I can use both skills. Software Engineering lets me turn my sense of color, design, and composition into real products like apps, websites, and games. I also enjoy solving problems step by step, which is exactly what programming asks for.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Software Developer, Front-End Developer, QA Tester	Writes and tests code under supervision; learns a programming language, Git, and teamwork; fixes bugs and builds small features.
Mid-Level	Software Developer, Full-Stack Developer, UI Engineer	Builds complete features on their own; works with databases and APIs; reviews code and writes tests.
Senior-Level	Senior Software Engineer, Technical Lead	Designs how the system is built, guides and mentors the team, makes technical decisions, and keeps code quality high.
Expert / Leadership	Software Architect, Engineering Manager, CTO	Sets the technical direction of products and the company, leads several teams, and connects technology to business goals.

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Programming (Python, JavaScript)	Y (basic)	Keep practicing with my CC 101 labs and build small programs every week.
Problem solving: algorithms, pseudocode, flowcharts	Y	Practice more problems and trace my programs by hand, like I did in my CC 101 activities.
Version control (Git and GitHub)	Y (just started)	I am using GitHub for my CC 101 portfolio; I will learn commits and branches by adding more projects.
UI/UX and visual design	Y	My art background gives me color, composition, and sketching; I will learn a design tool such as Figma.
Web development (HTML, CSS, JavaScript)	N	Follow free tutorials and build my own portfolio website to show my artwork.
Data structures and algorithms	N	Study it in my programming subjects and solve a few practice problems each week.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	
Company	

Item	Details
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	
Required Skills Listed (3–5)	

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose Software Engineering because it is the pathway where I can combine my art skills with computer science. Building apps, websites, and games needs both logic and creativity, and I enjoy both. The other four pathways focus more on servers, data, security, or business systems, which are less connected to design and visual work.

2. Looking at your roadmap, which career stage excites you the most, and why?

The mid-level stage excites me the most because I would be building complete features on my own, from the design to the working code. It feels like finishing a whole artwork instead of only a small part of it. I also look forward to growing toward senior roles where I can guide a creative team.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have basic programming, problem-solving, and design skills from my art background and my CC 101 activities. The hardest for me will probably be data structures, because it needs strong logic and a lot of practice. I plan to study it slowly and practice a few small problems every week.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Mapping this roadmap confirms that I should work hard in my programming subjects, not only in art. It also changed my plan a little: I now want to learn web development and Git early so I can build an online portfolio of my artwork. This gives me a clear goal for the rest of the course and my degree.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25

Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION
AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

Criteria	Description	Points
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date